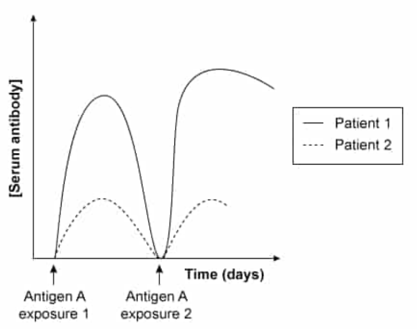
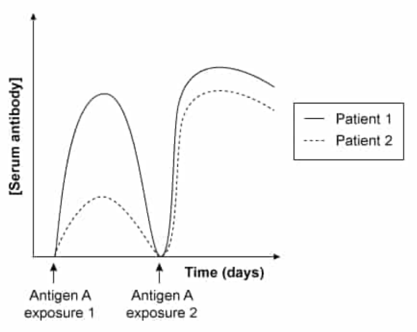
In a drug trial, one patient (Patient 1) initially received an injection containing inactivated antigen A, which can trigger an immune response. Another patient (Patient 2) initially received a saline injection. Neither patient had been previously exposed to antigen A. Following these initial injections, both patients were exposed to antigen A twice. Which of the following graphs shows the expected changes in the levels of serum antibodies against antigen A over time following antigen A exposures in these patients?

○ A.



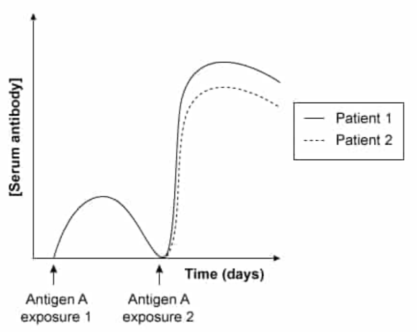
[9%]

✔ ◉ B.



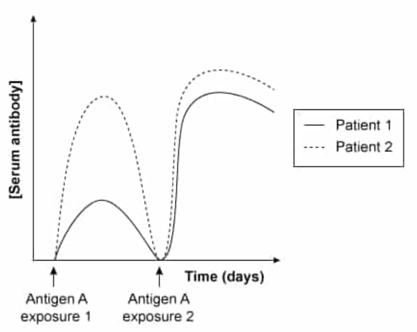
[67%]

○ C.



[15%]

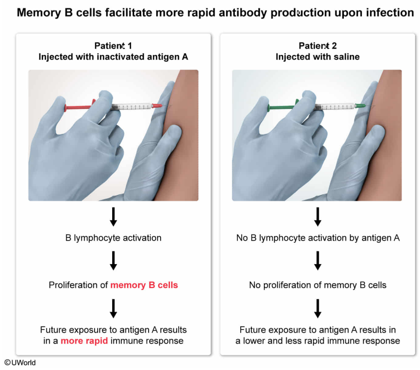
○ D.



[7%]

Correct
66% answered correctly

Explanation:



Cells of the adaptive immune system recognize specific foreign antigens and mount specialized immune responses against them. Specifically, B lymphocytes secrete antibodies that circulate throughout the blood and lymph and bind foreign antigens, marking them for destruction by other immune cells.

Upon binding a specific foreign antigen, a **B lymphocyte** becomes activated and engulfs (endocytoses) the foreign antigen, breaking it down into smaller fragments. Within the B lymphocyte, specialized antigen-binding peptides known as **major histocompatibility class II** (MHC II) **proteins** transport the antigen fragments to the cell surface, allowing the B lymphocyte to display the antigen fragments outside the cell.

Helper T cells bind foreign antigens displayed by B lymphocytes and release signaling molecules that stimulate B lymphocytes to divide and differentiate into both **short-lived, antibody-secreting plasma cells** and **long-lived memory B cells**. Memory B cells remain in lymphoid tissues for a long time and can recognize and respond to the foreign antigen **more rapidly** in the event of **a future infection**.

In this scenario, Patient 1 initially received an injection containing inactivated antigen A, which can trigger an immune response, and Patient 2 initially received a saline injection. Injection of inactivated antigen A into Patient 1 would have allowed B lymphocytes to bind the antigen, leading to the proliferation of both plasma cells and memory B cells. These **memory B cells** likely allowed a much more rapid response to the first post-injection Antigen A exposure, **increasing antibody production**. Upon the second post-injection exposure, memory B cells generated from either the initial injection or first post-injection exposure likely facilitated another rapid immune response.

In contrast, saline-treated Patient 2 was *not* initially exposed to inactive antigen A, and therefore did *not* initially produce memory B cells. Upon the first *post-injection* antigen A exposure, Patient 2 likely also produced antibodies against antigen A, but at a **lower** and **less rapid** rate than in Patient 1 **(Choices C and D)**. However, in Patient 2, this first-time antigen A exposure would have produced memory B cells that could then **more rapidly produce antibodies** following a **second exposure (Choice A)**. Only **Choice B** shows these results.

Educational objective:
Upon binding foreign antigens, activated B lymphocytes divide and differentiate into antibody-secreting plasma cells and long-lived memory B cells that can more rapidly respond to foreign antigens in the event of future infections.

Time spent:0 sec
Subject:
Biology

QID:402682
System:
Skin and Immune Systems